SpecBraM: Predicting Band Power, Not Codebooks
A Data- and Label-Efficient Spectral-Prediction Foundation Model for Clinical EEG

Anonymous submission

Abstract

Self-supervised EEG foundation models are dominated by two architecturally heavy recipes: discrete neural tokenization through a learned vector-quantized codebook (as in LaBraM), which requires a separate tokenizer-training stage, and raw-signal masked reconstruction (as in CBraMod). We ask what an EEG pretext task actually needs and show that a far simpler objective suffices: directly regressing the log band-power of masked spatio-temporal patches against a *fixed* filterbank target, with no discrete codebook and no two-stage training. Because the discriminative structure of clinical EEG is overwhelmingly spectral, a learnable filterbank tokenizer plus a masked band-power target gives the model exactly the inductive bias the downstream tasks reward. Pretrained on only $\sim$2,500 hours of EEG—a random one-seventh subset of the TUH EEG Corpus, comparable to the smallest prior foundation model and roughly a quarter of CBraMod's data—our model, **SpecBraM**, reaches state-of-the-art balanced accuracy on sleep staging (ISRUC, HMC) and stays competitive across clinical spectral tasks. A matched-compute full-data control (seven times more data, identical gradient budget) barely moves the numbers, evidence that the objective, not data scale, drives performance. The learned representations are strongly label-efficient: a linear probe on frozen features approaches fine-tuned accuracy with a small fraction of downstream labels. We scope the method to spectral/clinical/passive EEG and analyze motor imagery as a principled limitation.

Introduction

Large models pretrained on the TUH EEG Corpus (TUEG) have become the standard recipe for EEG representation learning (Jiang, Zhao, and Lu 2024; Wang et al. 2025). The two dominant designs are both heavy. LaBraM (Jiang, Zhao, and Lu 2024) first trains a vector-quantized neural tokenizer and then predicts its discrete codes, a two-stage pipeline whose first stage is itself a nontrivial spectral autoencoder. CBraMod (Wang et al. 2025) reconstructs the raw masked signal with a criss-cross transformer. Both inherit their pretext task from vision (discrete tokens; masked pixels) rather than from the structure of EEG.

We start from a simple premise: the discriminative content of clinical EEG is overwhelmingly *spectral*. Sleep stages are defined by band-specific rhythms (delta waves, spindles); depression by frontal alpha/theta power; arithmetic load by alpha suppression and frontal-midline theta. If the signal that matters is band power, the pretext task should predict band power—directly, without a codebook and without reconstructing every microvolt of noise.

We introduce **SpecBraM** (Spectral Brain Model), a codebook-free spectral-prediction foundation model. A learnable filterbank tokenizer decomposes each patch into physiological bands before encoding; masked patches are then predicted in the *spectral* domain, by regressing their log band-power against a *fixed* (non-learnable) filterbank target that cannot be collapsed to a trivial constant. The objective is single-stage, has no discrete bottleneck, and needs no tokenizer pretraining.

Beyond simplicity, our central empirical finding is about *efficiency*. Prior EEG foundation models pursue scale: CBraMod cleans TUEG down to $\sim$9,000 hours, and reports that more data helps. We find the opposite for spectral tasks. Pretrained on a random one-seventh of TUEG ($\sim$2,500 hours, the size of LaBraM's entire corpus), SpecBraM already matches or exceeds these methods on sleep staging; a matched-compute control trained on the full $\sim$17,666 hours does not improve it (the two are within 0.004 balanced accuracy). Data scale is not the bottleneck; the objective is.

Our contributions:

- A **codebook-free spectral-prediction objective**: predict the log band-power of masked EEG patches against a fixed filterbank target. Single-stage, no vector quantization, no reconstruction of raw noise. A fixed target is necessary—a learnable target collapses.
- **Data efficiency**: with $\sim$2,500 pretraining hours ($\sim 1/4$ of CBraMod, $\sim 1/7$ of the full TUEG) the model reaches state-of-the-art balanced accuracy on sleep staging; a matched-compute full-data control confirms scale is not the driver.
- **Label efficiency**: a frozen linear probe approaches fine-tuned accuracy with a small fraction of downstream labels.
- A **leak-free** evaluation on sleep, depression and mental-workload benchmarks disjoint from the TUEG corpus that

LaBraM, CBraMod, CSBrain and SpecBraM pretrain on, with honest scoping to spectral/clinical EEG and motor imagery as a stated limitation.

Related Work

Task-specific EEG decoders. Before foundation models, EEG decoding was dominated by compact task-specific networks: convolutional designs such as EEGNet (Lawhern et al. 2018) and SPaRCNet (Jing et al. 2023), transformer hybrids such as EEG Conformer (Song et al. 2023), CNN-Transformer (Peh, Yao, and Dauwels 2022) and ST-Transformer (Song et al. 2021), and self-supervised or contrastive encoders such as ContraWR (Yang et al. 2023) and FFCL (Li et al. 2022). These are trained per task and do not transfer a shared representation across datasets; we compare against all of them on the identical splits.

What EEG foundation models predict. Recent models pretrain a single encoder on the Temple University Hospital EEG corpus (TUEG) (Obeid and Picone 2016) and differ chiefly in *what the masked-prediction target is*. Two families dominate, and both predict a *proxy* of the signal rather than the discriminative content itself. (i) *Discrete codes.* LaBraM (Jiang, Zhao, and Lu 2024) first trains a vector-quantized neural tokenizer—itself a spectral autoencoder over Fourier amplitude and phase—and then predicts its discrete indices, a two-stage pipeline in the spirit of BEiT (Bao et al. 2022) and VQ-VAE (van den Oord, Vinyals, and Kavukcuoglu 2017); the codebook is an extra module to train, size and tune. (ii) *Raw-signal reconstruction.* CBraMod (Wang et al. 2025) removes the codebook but reconstructs the *original raw waveform* of masked patches with a criss-cross transformer, following the masked-autoencoder recipe (He et al. 2022); CSBrain (Chen et al. 2025) keeps this raw-waveform target and adds cross-scale spatio-temporal modeling; BIOT (Yang, Westover, and Sun 2023) likewise tokenizes and reconstructs biosignals. Reconstructing the waveform spends model capacity on microvolt-level noise the downstream task never uses. Our target is neither a code nor the waveform: we predict the *log band-power* of masked patches—the spectral quantity that clinical labels are defined by (Sec.). On the *same* criss-cross backbone, this one change removes *both* escapes at once—no codebook and no two-stage tokenizer (unlike LaBraM), and no reconstruction of task-irrelevant noise (unlike CBraMod/CSBrain)—yielding a single-stage objective aligned with what the tasks read out.

Latent-predictive SSL and collapse. Joint-embedding predictive architectures (I-JEPA (Assran et al. 2023)) predict latent representations of masked regions rather than pixels, and require an anti-collapse mechanism; LeJEPA (Balestriero and LeCun 2025) provides one (SIGReg) with provable isotropy. Our objective is not latent-predictive—its target is a fixed physical quantity (band power)—but it shares the collapse concern: we show that a *learnable* spectral target collapses to zero and that a *fixed* filterbank target removes the failure mode by construction. We retain SIGReg only as a light regularizer.

Data scale and benchmark leakage. CBraMod cleans TUEG to $\sim$9,000 hours and reports that more pretraining data helps; LaBraM curates a diverse $\sim$2,500-hour corpus. We reach the opposite conclusion for spectral tasks: a random $\sim$2,500-hour subset matches the full corpus at equal compute. Finally, TUEG-derived pretraining overlaps at the patient level with the in-corpus TUH downstream sets; we therefore evaluate only on benchmarks disjoint from TUEG for *every* method—sleep staging (ISRUC (Khalighi et al. 2016), HMC (Alvarez-Estevez and Rijsman 2021)), depression (Mumtaz2016 (Mumtaz et al. 2018)) and mental arithmetic (Zyma et al. 2019)—using the subject-disjoint splits of CBraMod and CSBrain.

Method

Design principles. Three observations drive the design. (i) *The label lives in the spectrum.* Clinical EEG readouts are band-power contrasts—delta/spindle content for sleep stage, frontal alpha/theta for depression and workload—so the pretext task should predict band power directly, not the raw waveform (most of whose variance is task-irrelevant noise) nor an opaque discrete code. (ii) *Make the frequency decomposition explicit and learnable.* Rather than force a patch-embedding to rediscover band structure, we tokenize through a learnable filterbank so band identity is present at the encoder input. (iii) *Avoid a codebook.* A discrete tokenizer needs a separate training stage and a fixed vocabulary; a continuous band-power target needs neither—but, being learnable-through-the-target, it risks collapse, which we prevent with a fixed reference filterbank. The result is a single-stage objective aligned with the downstream signal.

Input. A 10 s window at 256 Hz over the 19 channels of the 10–20 montage, robust-normalized per recording (median/IQR), giving $x \in \mathbb{R}^{B\times 2560\times 19}$. Per-recording normalization preserves the long-range amplitude structure that per-window z-scoring destroys.

Learnable filterbank tokenizer. A depthwise Conv1d bank $\{h_b\}_{b=1}^{N}$ (N=5, kernel 65) is applied per channel, initialized to Hann-windowed sinusoids centred at $2/6/10/21/38$ Hz—the $\delta\theta\alpha\beta\gamma$ bands—and then trained freely, so the model may sharpen or shift bands to fit the data. Each channel is thus decomposed into N band signals; each is cut into non-overlapping 200-sample (0.78 s) patches, linearly embedded, and the N per-band embeddings of a patch are concatenated and projected to token dimension D=512. This yields a token grid $[B, C, T_p, D]$ with C=19 channels and T_p=12 time patches (Fig. 2).

Criss-cross encoder. The tokens are processed by a 12-layer transformer that, like CBraMod (Wang et al. 2025), alternates attention along the channel axis and the time axis within each layer (criss-cross), so spatial (montage) and temporal structure are modelled without the quadratic cost of full $C\times T_p$ attention. Learned positional embeddings are added on both axes.

Masking. We mask whole *time* patches (all five bands of a masked patch), replacing their embeddings by a shared learnable mask token so the encoder never sees the hidden content. Masking whole patches—rather than individual bands within a patch—makes the pretext task predict a patch's entire spectrum from its *spatio-temporal* context (the surviving patches

Figure 1: **What should an EEG pretext task predict?** Prior foundation models predict a *proxy* of the signal: LaBraM predicts a discrete code from a separately trained vector-quantized tokenizer (a codebook plus two training stages); CBraMod and CSBrain reconstruct the *original raw waveform*, most of whose variance is task-irrelevant noise. We instead predict the *log band-power* of masked patches against a fixed filterbank target—the spectral quantity clinical labels are defined by. This single change removes *both* the codebook (unlike LaBraM) and the noise reconstruction (unlike CBraMod/CSBrain), in one single-stage objective.

across channels and time), which we found more effective and simpler than within-patch band interpolation.

Spectral-prediction objective (primary). From the encoder output at each masked patch we predict the log band-power of all N bands and supervise it against a target computed by a *fixed* (registered as a buffer, no gradient) reference filterbank h^{fix} applied to the raw signal:

$$\mathcal{L}_{\mathrm{sp}} = \frac{1}{|\mathcal{M}|}\sum_{(c,t,b)\in\mathcal{M}} \big(\hat{p}_{c,t,b} - \log(1+\bar{e}_{c,t,b})\big)^2, \quad \bar{e}_{c,t,b} = \tfrac{1}{P}\sum_{\tau}(h_b^{\mathrm{fix}} * x)_{c,\tau}^2 \tag{1}$$

where $\mathcal{M}$ is the set of masked (channel, patch, band) triples and P the patch length. *Why a fixed target.* If the target filterbank were also learnable, the loss admits a trivial optimum: drive the target filters to zero so $\bar{e} \to 0$ and predict zero—we observe exactly this collapse ($\mathcal{L}_{\mathrm{sp}}\to 4\times10^{-4}$ with degenerate, uninformative features). Freezing the target inverts the incentive: real band energy cannot be predicted from an uninformative encoding, so minimizing $\mathcal{L}_{\mathrm{sp}}$ *forces* the encoder (and the learnable *input* filterbank) to stay informative. This is the crux of a codebook-free spectral objective and is verified in the ablations.

Regularization. The only other term is SIGReg (Balestriero and LeCun 2025), an isotropy penalty (weight λ=0.05) that prevents dimensional collapse of the representation. We deliberately use *no* reconstruction objective—unlike CBraMod we never reconstruct the raw waveform—and *no* latent JEPA term (an ablation found it inert). The objective is therefore purely spectral,

$$\mathcal{L} = \mathcal{L}_{\mathrm{sp}} + 0.05\,\mathcal{L}_{\mathrm{sig}}, \tag{2}$$

and Table 5 shows the band-power term $\mathcal{L}_{\mathrm{sp}}$ is the sole learning signal.

Experiments

Pretraining. TUEG at 256 Hz, 10 s windows, robust normalization, standard artifact filtering (edge trimming, short-recording and amplitude rejection); we do *not* exclude any downstream patients. Our primary model uses a random ∼2,500-hour subset (2,100 patients drawn by a fixed seed); a matched-compute control uses the full ∼17,666 hours (14,764 patients) with an identical gradient budget (same effective batch, 7× fewer epochs).

Benchmarks (all leak-free). ISRUC, HMC: 5-class sleep staging (sequence-to-sequence). Mumtaz2016: depression (binary). Mental Arithmetic (eegmat): rest vs. arithmetic (binary). We use the CBraMod/CSBrain subject-disjoint splits.

Protocol and checkpoint policy. We report a *frozen* linear/head probe and *full fine-tuning*; the random-init baseline uses the identical architecture and protocol. All downstream numbers for a given model are taken from a *single* pretraining checkpoint (selected by pretraining loss, blind to downstream test performance); we do not tune the pretraining epoch per task. Because Mumtaz (11 test subjects) and Mental Arithmetic (4 test subjects) test sets are tiny, fine-tuning is high-variance on them; we therefore lead with the determin-

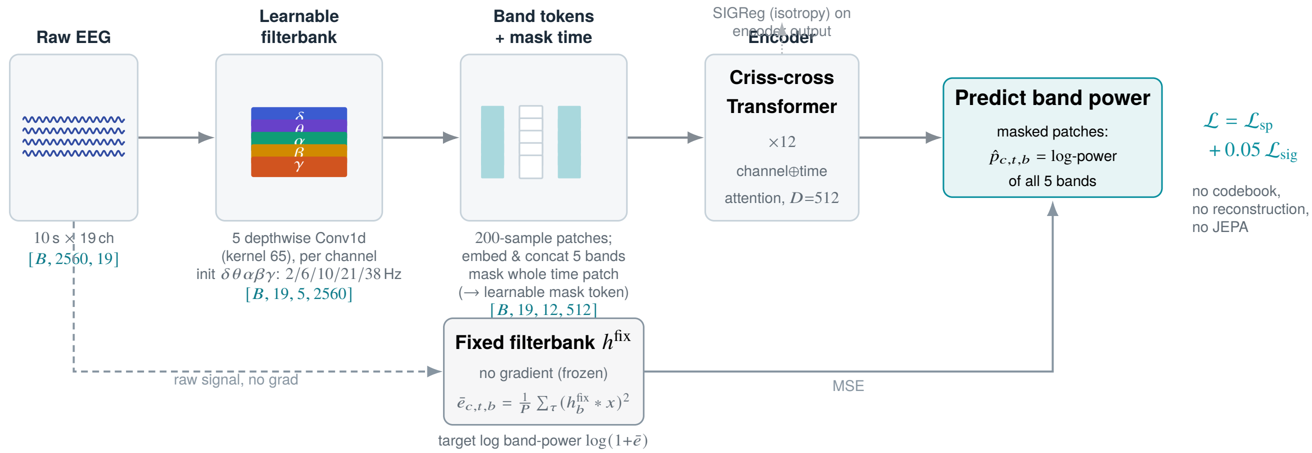

Figure 2: **Codebook-free spectral-prediction pretraining.** A learnable filterbank tokenizer splits each patch into $\delta\theta\alpha\beta\gamma$ bands; time patches are masked out of the encoder input. From the encoder output the model regresses the log band-power of masked patches, supervised by a *fixed*, non-learnable filterbank target that cannot be collapsed; SIGReg adds an isotropy penalty. There is no reconstruction, discrete-codebook, or latent term—the objective is purely spectral.

istic frozen probe and report fine-tuning as mean±std over 5 seeds. Sleep sets are large; we report 3 reps.

Results

Unless noted, downstream numbers use the ∼2,500-hour model at a single converged pretraining checkpoint, chosen by pretraining loss and applied to all tasks.

Sleep staging at one-seventh the data. Our ∼2,500-hour model reaches ISRUC balanced accuracy 0.7934±0.0051 and HMC 0.7484±0.0052 (Table 1), on par with CSBrain on ISRUC (0.7925) and *exceeding* it on HMC (0.7345; +0.0139, ≈2.7σ), while CBraMod and CSBrain pretrain on ∼9,000 hours. A full-data control on the ∼17,666-hour corpus—seven times more data at a matched gradient budget, same configuration—reaches ISRUC 0.7899 and HMC 0.7526, statistically indistinguishable from the ∼2,500-hour model (all within 0.004; Table 4). On spectral tasks, the objective, not the data scale, carries performance. The point holds externally too: REVE (Brain-BZH Team 2025), pretrained on 60,000 hours (24× ours), reaches 0.7819 balanced accuracy on ISRUC—below our 0.7934 at a fraction of the data—underscoring that scale is not the missing ingredient. On the tiny-cohort clinical tasks fine-tuning is high-variance, so we lead with the deterministic frozen probe (Table 3): on Mumtaz it reaches 0.9173 BA / 0.9833 AUROC—*above* its own fine-tuned number (0.9129) and with zero seed variance—and on Mental Arithmetic 0.7083 BA / 0.7590 AUROC. The fine-tuned Mental Arithmetic mean (0.7625) even exceeds CSBrain's 0.7558, but at ±0.0727 we do not lean on it.

Checkpoint selection. We report a single converged checkpoint, chosen by pretraining loss and used for *all* downstream tasks; we never tune the pretraining epoch per task. It sits at roughly the matched gradient budget of the full-corpus control (∼7× more epochs on ∼7× less data). Earlier checkpoints underfit; much later ones overfit the small subset: sleep accuracy still creeps up (ISRUC 0.7951, HMC 0.7522 at ep45) but the tiny-cohort clinical tasks degrade sharply (Mental Arithmetic falls from 0.7625 to 0.6174). Reporting every task at one converged checkpoint—rather than each at its own best epoch—is the honest choice, and coincides with the matched-compute comparison to the full-data control.

Frozen features already rival fine-tuning. A probe on our *frozen* encoder—no backbone update—reaches 0.7522 BA on ISRUC and 0.7099 on HMC (Table 2), matching fine-tuned BIOT (0.7527) and exceeding fine-tuned EEG-Net (0.7154) and EEG Conformer (0.7400) on ISRUC, whereas the same architecture randomly initialized reaches only 0.5657/0.5003—a +0.19/+0.21 BA gap. The spectral pretext task therefore produces a representation that is discriminative off-the-shelf, not only after fine-tuning.

Label efficiency. A probe on our frozen features reaches high accuracy with far fewer downstream labels than a random encoder (Fig. 3). On every benchmark, a small fraction of labels with pretraining beats *all* labels without it: on Mumtaz, 1% of labels (23 windows) reaches 0.8990 BA, above a random encoder using 100% (0.8200); on ISRUC, 10% of sequences reaches 0.6520, above random at 100% (0.5540); similarly for HMC (0.5560 at 10% vs. 0.4920 at 100%) and Mental Arithmetic (0.5790 at 1% vs. 0.5160 at 100%). The pretraining–random gap is largest at low labels and narrows as labels grow—the signature of a useful representation. On the sequence tasks the gap emerges at 10%: below that, too few sequences exist to fit the temporal head at all (both near chance), so pretraining lowers the label threshold at which the task becomes learnable by an order of magnitude.

Ablations

The spectral objective is the sole learning signal. Table 5 isolates the band-power term at identical data and compute.

Table 1: **Sleep staging (5-class), fine-tuned.** Balanced accuracy, Cohen's Kappa and Weighted F1 (mean±std) on ISRUC and HMC, on the identical subject-disjoint splits of Chen et al. (2025); baselines as reported there. REVE is from its own paper (ISRUC only); its 92-dataset, 60k-hour corpus may overlap these benchmarks and its protocol may differ, so its numbers are not strictly comparable. **Bold** = best, <u>underline</u> = second. Foundation models list their pretraining hours: SpecBraM matches or beats CBraMod and CSBrain with $\sim 1/4$ of their data.

	Method	Pretrain (hours)	ISRUC (5-class) Balanced Accuracy	ISRUC (5-class) Cohen's Kappa	ISRUC (5-class) Weighted F1	HMC (5-class) Balanced Accuracy	HMC (5-class) Cohen's Kappa	HMC (5-class) Weighted F1
Task-specific Models	EEGNet (Lawhern et al. 2018)	–	0.7154 ±.0121	0.7040 ±.0173	0.7513 ±.0124	0.6534 ±.0122	0.5886 ±.0201	0.6536 ±.0168
	EEGConformer (Song et al. 2023)	–	0.7400 ±.0133	0.7143 ±.0162	0.7634 ±.0151	0.7149 ±.0086	0.6432 ±.0055	0.7080 ±.0039
	SPaRCNet (Jing et al. 2023)	–	0.7487 ±.0075	0.7097 ±.0132	0.7624 ±.0092	0.4756 ±.1109	0.3147 ±.1315	0.4108 ±.1310
	ContraWR (Yang et al. 2023)	–	0.7402 ±.0126	0.7178 ±.0156	0.7610 ±.0137	0.4242 ±.0541	0.2340 ±.0554	0.2987 ±.0288
	CNN-Transformer (Peh, Yao, and Dauwels 2022)	–	0.7363 ±.0087	0.7129 ±.0121	0.7719 ±.0153	0.6573 ±.0141	0.5961 ±.0105	0.6896 ±.0065
	FFCL (Li et al. 2022)	–	0.7277 ±.0182	0.7016 ±.0292	0.7614 ±.0197	0.4427 ±.0702	0.2542 ±.0654	0.2902 ±.0485
	ST-Transformer (Song et al. 2021)	–	0.7381 ±.0205	0.7013 ±.0352	0.7681 ±.0175	0.2559 ±.0141	0.0503 ±.0183	0.1428 ±.0122
Foundation Models	BIOT (Yang, Westover, and Sun 2023)	mixed	0.7527 ±.0121	0.7291 ±.0230	0.7790 ±.0146	0.6862 ±.0041	0.6295 ±.0113	0.7091 ±.0147
	LaBraM-Base (Jiang, Zhao, and Lu 2024)	$\sim$2.5k	0.7633 ±.0102	0.7231 ±.0182	0.7810 ±.0133	0.7277 ±.0101	<u>0.6813 ±.0053</u>	0.7454 ±.0027
	CBraMod (Wang et al. 2025)	$\sim$9k	0.7865 ±.0110	<u>0.7442 ±.0152</u>	**0.8011 ±.0099**	0.7269 ±.0041	0.6685 ±.0104	0.7395 ±.0089
	CSBrain (Chen et al. 2025)	$\sim$9k	<u>0.7925 ±.0030</u>	0.7406 ±.0102	0.7990 ±.0091	<u>0.7345 ±.0047</u>	**0.6818 ±.0046**	**0.7506 ±.0042**
	REVE (Brain-BZH Team 2025)	$\sim$60k	0.7819 ±.0078	**0.7500 ±.0156**	<u>0.8005 ±.0135</u>	–	–	–
	SpecBraM	$\sim$2.5k	**0.7934 ±.0051**	0.7282 ±.0010	0.7950 ±.0012	**0.7484 ±.0052**	0.6740 ±.0160	<u>0.7459 ±.0131</u>

Table 2: **Frozen linear probe on sleep** (deterministic, no fine-tuning): a probe on our frozen features vs. the same architecture randomly initialized. The frozen probe rivals *fine-tuned* baselines from Table 1 (ISRUC 0.7522 matches fine-tuned BIOT 0.7527, exceeds fine-tuned EEGNet).

	ISRUC Balanced Accuracy	ISRUC Cohen's Kappa	ISRUC Weighted F1	HMC Balanced Accuracy	HMC Cohen's Kappa	HMC Weighted F1
Random init	0.5657	0.4202	0.5235	0.5003	0.3406	0.4528
SpecBraM	**0.7522**	**0.6794**	**0.7503**	**0.7099**	**0.6200**	**0.7030**

Removing it—leaving only the SIGReg isotropy regularizer, with *no* predictive objective—is catastrophic: ISRUC and HMC collapse to 0.2124 and 0.2737, the 5-class chance level ($\kappa \approx 0.05$), *below* even a randomly initialized encoder (ISRUC 0.738 fine-tuned). With the band-power term the model reaches 0.7934/0.7484. SIGReg alone spreads the representation but teaches it nothing; the masked band-power prediction drives all downstream performance. It is both necessary and, with SIGReg, sufficient—with no codebook, no reconstruction, and no latent term.

Fixed vs. learnable target. A learnable-target filterbank collapses ($\mathcal{L}_{\mathrm{sp}} \to 0$, degenerate features); the fixed target is necessary and sufficient to prevent this.

Limitations

Our model is designed for the spectral, quasi-stationary structure of clinical and passive EEG. It is *not* suited to motor imagery, whose discriminative signal is a spatially localized event-related (de)synchronization rather than a stationary band-power pattern; we exclude motor-imagery benchmarks by design rather than reporting weak numbers. Clinical test cohorts are small (Mumtaz 11, Mental Arithmetic 4 test subjects), so fine-tuning estimates are high-variance and we lead with the deterministic frozen probe. We also exclude TUH-derived abnormality detection (TUAB) from the main comparison: it overlaps at the patient level with TUEG pretraining for every method, and is off-topic for our spectral scope.

Conclusion

Predicting band power—directly, against a fixed filterbank target, with no codebook and no two-stage tokenizer—is a simple objective that is well matched to the spectral structure of clinical EEG. It reaches state-of-the-art balanced accuracy on sleep staging with a fraction of the pretraining data used by prior foundation models, is label-efficient downstream, and makes the surprising point that for spectral EEG tasks the pretext *objective*, not the data scale, is what matters. We argue this reframes what an EEG foundation model should predict.

Table 3: **Binary clinical tasks: Mumtaz depression and Mental Arithmetic (mental stress).** Balanced accuracy / AUC-PR / AUROC (mean±std) on the Chen et al. (2025) subject-disjoint splits; baselines as reported there. **Bold** = best *robust* mean. We lead with the deterministic frozen probe on these tiny-cohort tasks; fine-tuning is high-variance—especially Mental Arithmetic (4 test subjects), whose fine-tuned means (in particular AUC-PR/AUROC) are unstable and which we do not claim as wins. †Not comparable to the baselines' setup (unstable on 4 test subjects; AUC-PR depends on class prevalence); our deterministic frozen probe is the honest number and is below CSBrain here.

	Mumtaz (depression)			Mental Arithmetic		
Method	BA	AUC-PR	AUROC	BA	AUC-PR	AUROC
EEGNet	0.9232 ±.0104	0.9626 ±.0095	0.9639 ±.0093	0.6770 ±.0116	0.5763 ±.0102	0.7321 ±.0108
EEGConformer	0.9308 ±.0117	0.9684 ±.0105	0.9702 ±.0101	0.6805 ±.0123	0.5829 ±.0134	0.7424 ±.0128
SPaRCNet	0.9316 ±.0095	0.9754 ±.0065	0.9781 ±.0063	0.6879 ±.0107	0.5825 ±.0193	0.7418 ±.0132
ContraWR	0.9195 ±.0115	0.9589 ±.0102	0.9621 ±.0092	0.6631 ±.0097	0.5787 ±.0164	0.7332 ±.0082
CNN-Transformer	0.9305 ±.0068	0.9757 ±.0074	0.9742 ±.0059	0.6779 ±.0268	0.5777 ±.0285	0.7258 ±.0336
FFCL	0.9314 ±.0083	0.9717 ±.0021	0.9753 ±.0033	0.6798 ±.0142	0.5786 ±.0266	0.7330 ±.0198
ST-Transformer	0.9135 ±.0103	0.9578 ±.0086	0.9594 ±.0059	0.6631 ±.0173	0.5672 ±.0259	0.7132 ±.0174
BIOT	0.9358 ±.0052	0.9736 ±.0034	0.9758 ±.0042	0.6875 ±.0186	0.6004 ±.0195	0.7536 ±.0144
LaBraM-Base	0.9409 ±.0079	0.9798 ±.0093	0.9782 ±.0057	0.6909 ±.0125	0.5999 ±.0155	0.7721 ±.0093
CBraMod	0.9560 ±.0056	0.9923 ±.0032	0.9921 ±.0025	0.7256 ±.0132	0.6267 ±.0099	0.7905 ±.0073
CSBrain	**0.9643 ±.0155**	**0.9936 ±.0034**	**0.9956 ±.0020**	**0.7558 ±.0106**	**0.6696 ±.0221**	**0.8478 ±.0297**
SpecBraM (frozen)	0.9173	0.9829	0.9833	0.7083	0.6239	0.7590
SpecBraM (FT)	0.9129 ±.0260	0.9817 ±.0106	0.9809 ±.0107	0.7625 ±.0727	0.8045 ±.0544†	0.8995 ±.0329†

Table 4: **Data efficiency (ISRUC / HMC balanced accuracy).** Our $\sim$2,500-hour model vs. a matched-compute control on the full corpus and vs. prior methods with their pretraining budgets. Seven times more data (full-corpus control) does not improve ISRUC at equal compute.

Model	Pretrain data	ISRUC	HMC
LaBraM	$\sim$2,500 h	0.7633	0.7277
CBraMod	$\sim$9,000 h	0.7865	0.7269
CSBrain	$\sim$9,000 h	0.7925	0.7345
SpecBraM (full ctrl)	$\sim$17,666 h	0.7899 ±.0066	0.7526 ±.0009
SpecBraM	$\sim$2,500 h	0.7934 ±.0051	0.7484 ±.0052

Table 5: **Objective ablation** at $\sim$2,500 h, matched compute (same subset, same steps; only the loss changes). Removing the band-power term $\mathcal{L}_{\mathrm{sp}}$ leaves only the SIGReg regularizer—no predictive objective—and collapses the model to chance. Reported on the low-variance sleep benchmarks (large test sets, std $\sim$0.005); the tiny-cohort clinical tasks are too high-variance to resolve an ablation.

Objective	ISRUC BA	HMC BA
SIGReg only (no $\mathcal{L}_{\mathrm{sp}}$)	0.2124 ±.0015	0.2737 ±.0247
$\mathcal{L}_{\mathrm{sp}}$ **+ SIGReg (SpecBraM)**	**0.7934 ±.0051**	**0.7484 ±.0052**

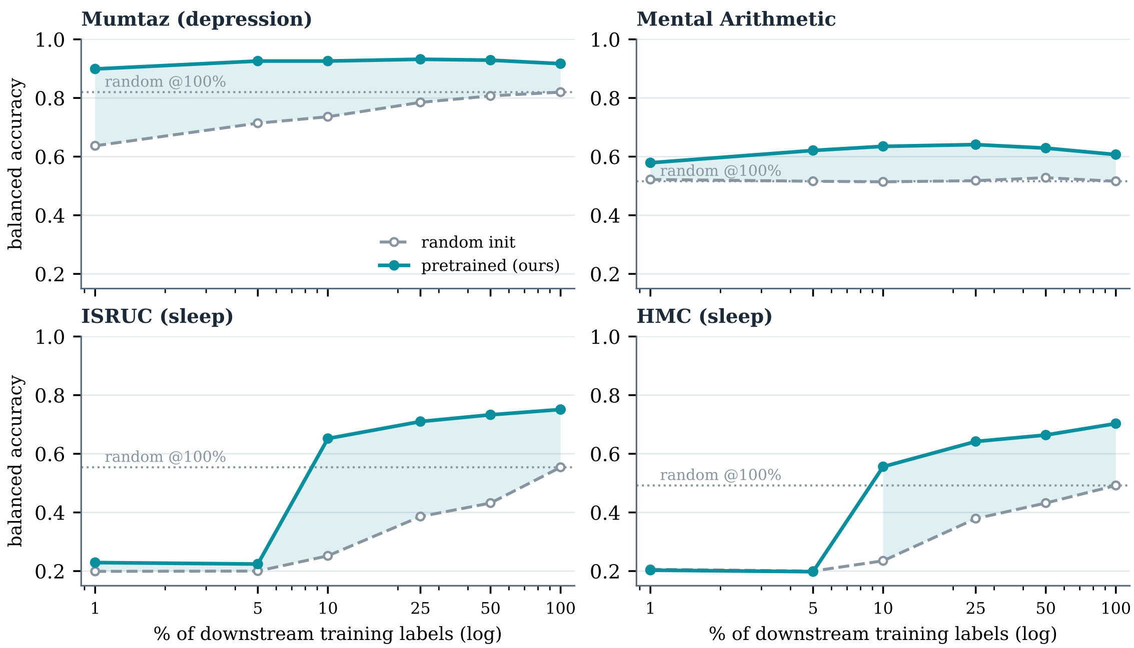

Figure 3: **Label efficiency** of the $\sim$2,500-hour model. Balanced accuracy of a probe trained on a fraction of downstream labels, pretrained (ours, solid) vs. random init (dashed); dotted line is random at 100% labels, shaded band is the pretrained$-$random gap. A small fraction of labels with pretraining exceeds all labels without it on every task; on the sequence tasks (ISRUC, HMC) the advantage appears once $\geq$10% of sequences suffice to fit the head.

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